

Congruences for an Iterative Generating Function: A Proof of the A396798 Iteration Conjectures

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Abstract

Let $A(x) \in x + x^2\mathbb{Z}[[x]]$ be determined by $A(x) = x + A^{\circ 4}(x)A^{\circ 5}(x)$, where superscripts with a composition symbol denote iteration. We prove that, for every integer k ,

$$A^{\circ k}(x) \equiv \frac{x}{1-kx} + 4\varepsilon_k \frac{x^4}{(1-x)^3} \pmod{8}, \quad \varepsilon_k \in \{0, 1\}, \quad \varepsilon_k \equiv k \pmod{2}.$$

This establishes the seven iteration congruences listed as conjectures in OEIS A396798 at the time of consultation. It also gives the precise starting index of the coefficient pattern for A itself. The proof uses uniqueness of a formal fixed point and an explicit rational function over $\mathbb{Z}/8\mathbb{Z}$. In this ring the reduced series has compositional order exactly eight. An accompanying dependency-free Python program checks the rational identities exactly and independently reconstructs initial coefficients.

1 The problem and its documented status

Paul D. Hanna's sequence A396798 is defined in OEIS by the ordinary generating function

$$A(x) = \sum_{n \geq 1} a(n)x^n = x + A^{\circ 4}(x)A^{\circ 5}(x). \tag{1}$$

Thus $a(1), a(2), \dots$ begin 1, 1, 9, 133, 2517, 55609. The notation $A^{\circ j}$ means j successive substitutions of A , not the ordinary product power $A(x)^j$. We set $A^{\circ 0}(x) = x$. Negative iterates use the compositional inverse, which exists because the coefficient of x is one.

The source record [1], checked on September 22, 2026, labels seven assertions about iterates 2 through 8 as conjectures. They predict vanishing coefficient tails for even iterates and periodic residue blocks for odd iterates. The result below proves all seven. The record also proposes a residue block for $a(n)$ itself; its stated starting index needs the correction explained in Section 5. This is a specific set of published sequence conjectures. The source check documents their listed status; it does not certify that no unindexed proof exists elsewhere.

Throughout, congruences of formal series mean coefficientwise congruences. No analytic convergence is required. Put $R = \mathbb{Z}/8\mathbb{Z}$ and define

$$M_k(x) = \frac{x}{1-kx}, \quad H(x) = \frac{x^4}{(1-x)^3}, \quad F(x) = M_1(x) + 4H(x). \tag{2}$$

Every denominator here has constant coefficient one and is therefore invertible in the relevant formal power series ring.

Theorem 1. *The defining equation (1) has a unique solution in $x + x^2\mathbb{Z}[[x]]$. For this solution and every $k \in \mathbb{Z}$,*

$$A^{\circ k}(x) \equiv M_k(x) + 4\varepsilon_k H(x) \pmod{8}, \quad (3)$$

where ε_k is the parity of k . In particular, for $n \geq 2$,

$$[x^n] A^{\circ k}(x) \equiv k^{n-1} + 4\varepsilon_k \binom{n-2}{2} \pmod{8}, \quad (4)$$

where $\binom{0}{2} = \binom{1}{2} = 0$. The coefficient of x is one for every k .

2 Uniqueness of the defining formal series

The following argument works over a commutative ring, including rings with zero divisors. This matters when reducing modulo eight.

Lemma 2. *Let S be a commutative ring with identity and let $r, s \geq 1$ be integers. There is exactly one $B \in x + x^2S[[x]]$ satisfying $B = x + B^{\circ r}B^{\circ s}$.*

Proof. For $B \in x + x^2S[[x]]$ write $T(B) = x + B^{\circ r}B^{\circ s}$. Suppose $B - C \in x^N S[[x]]$, with $N \geq 2$. Substitution by a series of order one preserves divisibility by x^N . Also, if $U - V$ is divisible by x^N , then so is $D(U) - D(V)$ for every formal series D for which the substitutions are defined: apply $U^j - V^j = (U - V) \sum_{i=0}^{j-1} U^{j-1-i} V^i$ term by term. Induction on the number of substitutions consequently gives $B^{\circ j} - C^{\circ j} \in x^N S[[x]]$ for all $j \geq 0$.

Both factors in the product defining T have order one, so

$$\begin{aligned} T(B) - T(C) &= (B^{\circ r} - C^{\circ r})B^{\circ s} + C^{\circ r}(B^{\circ s} - C^{\circ s}) \\ &\in x^{N+1}S[[x]]. \end{aligned}$$

Starting with $B_0 = x$ and putting $B_{j+1} = T(B_j)$, we obtain $B_{j+1} - B_j \in x^{j+2}S[[x]]$. Hence each coefficient eventually stabilizes. Their coefficientwise limit is a fixed point because addition, multiplication, and these substitutions respect finite truncations.

If two fixed points first differed in degree N , their images under T could not differ before degree $N+1$, a contradiction. This proves uniqueness. The construction also shows that reducing coefficients modulo an integer commutes with taking this fixed point. $\square$

3 An explicit series of order eight

The elementary fractional-linear identity

$$M_u(M_v(x)) = M_{u+v}(x) \quad (5)$$

holds over any commutative ring. The additional term $4H$ in F has a useful property: products of two such perturbations vanish in R .

Lemma 3. *In $R[[x]]$, $F^{\circ 2}(x) = M_2(x)$.*

Proof. Since $4^2 = 0$ in R , the binomial theorem gives the exact first-order substitution rule

$$P(U + 4V) = P(U) + 4VP'(U) \quad (6)$$

for formal series $U, V \in xR[[x]]$. There are no divisions by factorials in this identity. Apply it to $F = M_1 + 4H$:

$$F^{\circ 2} = M_1 \circ M_1 + 4(H \circ M_1 + (M_1' \circ M_1)H).$$

It is enough to evaluate the parenthesis modulo two. In $\mathbb{F}_2[[x]]$, direct substitution gives

$$H(M_1(x)) = \frac{x^4}{1-x}, \quad M_1'(M_1(x))H(x) = \frac{x^4}{1-x}.$$

Their sum is zero. The correction multiplied by four therefore vanishes modulo eight, and (5) proves the claim. $\square$

Lemma 4. *For every integer k , the identity $F^{\circ k} = M_k + 4\varepsilon_k H$ holds in $R[[x]]$.*

Proof. First take $k \geq 0$. By Lemma 3 and (5), $F^{\circ 2j} = M_{2j}$. For the next iterate,

$$F^{\circ 2j+1} = F \circ M_{2j} = M_{2j+1} + 4H \circ M_{2j}.$$

But $M_{2j}(x) \equiv x \pmod{2}$, so $4H \circ M_{2j} = 4H$ in $R[[x]]$. This proves the formula for nonnegative k .

In particular, $F^{\circ 8} = M_8 = x$. Thus F has an inverse under composition and its integer iterates depend only on k modulo eight. The right-hand side $M_k + 4\varepsilon_k H$ has the same dependence. The formula consequently holds for negative k as well. $\square$

4 Verification of the fixed point

We now identify F with the reduction of A . This step verifies the defining equation itself, rather than merely matching initial coefficients.

Lemma 5. *The series F satisfies $F = x + F^{\circ 4}F^{\circ 5}$ in $R[[x]]$.*

Proof. Lemma 4 gives $F^{\circ 4} = M_4 = x + 4x^2$ and $F^{\circ 5} = M_5 + 4H$ in $R[[x]]$. Consequently,

$$F^{\circ 4}F^{\circ 5} = \frac{x^2(1+4x)}{1-5x} + 4xH.$$

Using $F - x = x^2/(1-x) + 4H$, their difference is

$$\begin{aligned} F - x - F^{\circ 4}F^{\circ 5} &= \frac{x^2}{1-x} - \frac{x^2(1+4x)}{1-5x} + 4(1-x)H \\ &= 4x^4 \left(\frac{1}{(1-x)(1-5x)} + \frac{1}{(1-x)^2} \right) \quad \text{in } R[[x]]. \end{aligned}$$

The expression in parentheses is zero modulo two, because $1 - 5x \equiv 1 - x \pmod{2}$. After multiplication by four it is zero modulo eight. This proves the identity. $\square$

Proof of Theorem 1. Apply Lemma 2 over $\mathbb{Z}$ to obtain A , and over R to its reduction. Lemma 5 shows that F is the latter fixed point. Uniqueness therefore gives $A \equiv F \pmod{8}$. Composition, including composition with inverses, respects coefficient reduction. Lemma 4 proves (3).

Finally, $M_k(x) = \sum_{n \geq 1} k^{n-1} x^n$ and $H(x) = \sum_{n \geq 4} \binom{n-2}{2} x^n$. Extracting coefficients gives (4) and the separate coefficient-one statement in degree one. For $k = 0$, one may simply use $M_0 = x$, avoiding any need to interpret 0^0 . $\square$

5 The conjectures and the indexing correction

For odd k , the coefficient tail beginning at $n = 2$ repeats the block $(k, 1, k + 4, 5)$ modulo eight. Indeed, $k^2 \equiv 1 \pmod{8}$, while the parity of $\binom{n-2}{2}$ repeats 0, 0, 1, 1 from $n = 2$ onward. For even k , the perturbation vanishes and $k^3 \equiv 0 \pmod{8}$. The complete results for one period of iterates are as follows.

Iterate	Exact description modulo 8
1	Coefficient of x : 1; from $n = 2$: (1, 1, 5, 5) repeating
2	$x + 2x^2 + 4x^3$
3	Coefficient of x : 1; from $n = 2$: (3, 1, 7, 5) repeating
4	$x + 4x^2$
5	Coefficient of x : 1; from $n = 2$: (5, 1, 1, 5) repeating
6	$x + 6x^2 + 4x^3$
7	Coefficient of x : 1; from $n = 2$: (7, 1, 3, 5) repeating
8	x

Rows 2 through 8 prove the seven iteration assertions in the source. For the remaining coefficient claim, the repeating block (1, 1, 5, 5) starts at $n = 2$, with $a(1) = 1$ separate. Starting that same block at $n = 1$ would predict $a(3) \equiv 5$, whereas $a(3) = 9 \equiv 1$. Thus the literal starting index in the consulted record cannot be retained. The corrected formula, valid for every index, is

$$a(1) = 1, \quad a(n) \equiv \begin{cases} 1 & n \equiv 2, 3 \pmod{4}, \\ 5 & n \equiv 0, 1 \pmod{4} \end{cases} \quad (n \geq 2).$$

Corollary 6. *The reduction of A modulo eight has compositional order exactly eight. For all $k \in \mathbb{Z}$, $A^{\circ k+8} \equiv A^{\circ k} \pmod{8}$.*

Proof. The eighth iterate is x . Conversely, the coefficient of x^2 in $A^{\circ k}$ is k modulo eight, so an iterate equal to x must have $8 \mid k$. $\square$

6 Reproducible exact checks

The accompanying file `verify_a396798.py` uses Python's standard library only. It verifies rational functions by cross-multiplying their polynomial numerators and denominators in $(\mathbb{Z}/8\mathbb{Z})[x]$. All denominators have odd constant coefficients, so these checks certify identities of formal series to every degree. In particular, the script checks $F \circ F = M_2$, each transition between the eight proposed iterate formulas, and the defining identity $F = x + F^{\circ 4} F^{\circ 5}$.

As separate arithmetic checks, it reconstructs the first twenty integer coefficients directly from (1) and compares them with the source data. It then reconstructs A modulo eight through degree 64 and checks every coefficient of iterates $k = 0, \dots, 16$ against the theorem. All checks pass. These finite comparisons supplement the formal proof; the rational identities and the uniqueness argument supply the statements for arbitrary degree.

Run the checks with `python3 verify_a396798.py` in GITHUB https://github.com/PolloXDDD/a396798_repository. The manuscript is self-contained and compiles with `pdflatex`; no external figures or bibliography files are required.

References

- [1] Paul D. Hanna, *A396798: G.f. satisfies $A(x) = x + A^4(x)A^5(x)$* , The On-Line Encyclopedia of Integer Sequences, June 16, 2026. Here the superscripts in the source title denote iteration. <https://oeis.org/A396798>; internal record: <https://oeis.org/A396798/internal>. Consulted September 22, 2026.